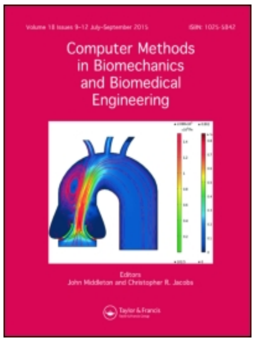

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Glottis effects on the cough clearance process simulated with a CFD dynamic mesh and Eulerian wall film model

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ABSTRACT

In this study, we have reproduced the cough clearance process with an Eulerian wall film model. The simulated domain is based on realistic geometry from the literature, which has been improved by adding the glottis and epiglottis. The vocal fold movement has been included due to the dynamic mesh method, considering different abduction and adduction angles and velocities. The proposed methodology captures the deformation of the flexible tissue, considers non-Newtonian properties for the mucus, and enables us to reproduce a single cough or a cough epoch. The cough efficiency (CE) has been used to quantify the overall performance of the cough, considering many different boundary conditions, for the analysis of the glottis effect. It was observed that a viscous shear force is the main mechanism in the cough clearance process, while the glottis closure time and the epiglottis position do not have a significant effect on the CE. The cough assistance devices improve the CE, and the enhancement rate grows logarithmically with the operating pressure. The cough can achieve an effective mucus clearance process, even with a fixed glottis. Nevertheless, the glottis closure substantially improves the CE results.

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KEYWORDS
Glottis; cough clearance; Eulerian wall film; CFD; epiglottis

1. Introduction

The cough is an innate reflex that ensures the removal of mucus and noxious substances, and it protects airways against infections. It is a natural defensive mechanism that is experienced by all people. A cough is the most common complaint for which individuals worldwide seek medical attention (Paul et al. 2014). The heath care cost associated with coughing is enormous; antitussive drugs already constitute an expense of more than 100 million pounds every year in the U.K. (Pandya et al. 2016) or billions of dollars in over-the-counter cough drugs in the United States alone (Dicpinigaitis 2015). The vast majority of cases of acute cough are due to the common cold, and this cough is typically transient and self-limited (Dicpinigaitis 2015). However, the reasons for coughing include bronchiectasis, laryngeal hypersensitivity (Hull and Menon 2015), chronic bronchitis, asthma, eosinophilic bronchitis and immune deficiency (Martin and Harrison 2015).

A cough can be initiated voluntary or by the stimulation of airway receptors. Inflammation and infection of the larynx, trachea and lower airways usually acts as a stimulus. The tussigenic stimulus is received by the brainstem, which initiates the cough reflex. After the start of the process, at least three more stages are usually distinguished, i.e. the inspiratory, compressive, and expulsive phases. In the inspiratory phase, the lung volume is expanded to produce sufficient expiratory power, and the air volume that is inspired ranges from 50% of the tidal volume to 50% of the vital capacity (McCool 2006). This action is followed by a brief compressive phase, approximately 200 ms for healthy subjects (McCool 2006), although this phase can be extended considerably due to diseases as congenital central hypoventilation syndrome (Lavorini et al. 2007). In this phase, the laryngeal adductor muscles narrow or close the glottis completely; there is no flow, and the expiration muscles contract forcefully, which allows the subglottic pressure to build up. Thus, the expulsive phase follows when the glottis is suddenly reopened, while the expiratory muscles continue with a forceful contraction, producing the characteristic sound. The expulsive phase can be interrupted by glottis closures into a cough epoch, which consists of consecutive compressive and expulsive phases without intermediate inspirations. This phase is crucial in the cough clearance process; the cough is ineffective if this phase fails. In healthy individuals, the mucus clearance should be completed only with mucociliary clearance (Lippmann et al. 1980); however, it should be remarked that the second mechanism for the expulsion

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of mucus from the airways is the cough clearance (Basser et al. 1989; Vasquez and Forest 2015).

Some studies point out that the glottis closure is an essential and definitive component of the cough (Chung and Mazzone 2016). It is a prerequisite for development of the high intra-thoracic pressure of the cough, and thus, it is necessary for the subsequent phases (Finder 2010). At least, the glottis closure is an important contributing factor (McCulloch et al. 2011; Choi et al. 2012). Fontana (2008) discusses the importance of the glottis in the cough and concludes that glottal closure would not necessarily be considered to be a part of the cough process. Nevertheless, other researches, (Gal 1980; Young et al. 1987) point out that this necessity is not very clear, and they found that the role of the glottis in the cough is questionable, by analysing cough efforts prior to and following tracheal intubation and obtaining similar values. McCool (2006) concludes that although the glottal closure enhances the compression phase, it is not essential for an effective cough. There appears to be general agreement that coughing requires the control and coordination of the glottis and the laryngeal muscles' motion and activity (Fontana and Lavorini 2006). When the expiratory muscles contract against a closed glottis, the developed pressure is approximately 50 to 100% greater than that obtained during other forced expiratory manoeuvres in which the glottis is open (McCool and Leith 1987; Lee et al. 2009).

The larynx physiology plays a fundamental role in the cough, and it involves an anatomically complex region (Thurnher et al. 2007). The epiglottis and the glottis along with the vocal fold are the most important parts of the larynx in cough research. Vocal fold diseases can affect the compression phase, reducing the cough flow peak and increasing the risk of respiratory complications (Ruddy et al. 2014). Certain dysfunctions are usually associated with cough symptoms (Cobeta et al. 2013; Dunn et al. 2015). Cough requires a rapid vocal fold adduction (closure) before the build-up of subglottal pressure, and it is followed by a rapid vocal fold abduction (opening) with the expulsion of air. These abduction and adduction processes of vocal fold coughing have broader ranges (Magni et al. 2011). There are mechanical devices that attempt to substitute dysfunctional glottises, such as an artificial external glottis device (AEGD) (Kim et al. 2011).

Quantification of the cough severity is a difficult task, and there are graphical systems (Birring and Spinou 2015; Dicpinigaitis 2015), but it is currently almost impossible to measure the cough clearance process *in vivo*. In this context, Computational Fluid Dynamics (CFD) techniques are emerging as tools for evaluating all types of flows that are related to the respiratory system. Experimentally validated results (Holbrook and Longest 2013) show that the airflow in the airways can be simulated by CFD techniques accurately. Other simulation studies have also been performed, for example, on drug administration (Oliveira et al. 2012; Park 2016), glottal motion on breathing (Scheinherr et al. 2012, 2015), or particle deposition (Inthavong et al. 2011). There are also detailed published studies on cough, but they are usually written from a medical point of view, such as Hilton et al. (2015). The cough clearance process is a multiphase flow. In general, it is difficult to approach it with classical CFD simulation models (Smith et al. 2008). The Eulerian Wall Film (EWF) model, developed for thin films adhered to walls, has been employed to quantify and evaluate the cough clearance effectiveness by Paz et al. (2017). This research attempts to reproduce the unsteady process and includes dynamic mesh methodology to simulate the deformation of the flexible tissue. However, the epiglottis and glottis presence is not considered, and thus, the movement of the glottis and the pressure effects on the clearance process have not been accounted for.

To our knowledge, there are no previous research studies on reproducing the unsteady mucus cough clearance process while considering the glottis action. In the present research, the cough clearance process has been analysed with Fluent's EWF model, based on the previous methodology published by Paz et al. (2017). The main physical considerations, such as the airway deformations, viscous mucus properties, and transient simulation, have been maintained. Moreover, a more realistic larynx geometry, with the glottis and epiglottis, has been considered. Also the vocal fold movement is included. This action has allowed us to analyse the impact of the glottis movement on the cough clearance. Differences between a single cough and a cough epoch and the effect of cough assistance devices on the clearance process have also been studied. In addition, the contributions of each of the mechanisms involved in the cough clearance process have been compared. The main aim of this research is to help to elucidate the importance of the glottis in the cough.

2. Methodology

2.1. Geometry and mesh

Many previous cough research studies have confirmed that most of the airflow in a cough, or even all of the airflow, flows through the mouth, and thus, it is possible to use a simplified model (Johari et al. 2013). In this study, the model corresponds to the upper airway form the trachea to the mouth, as taken from the Paz et al. research (Paz et al. 2017); however, their larynx was improved including the glottis and epiglottis, to produce a more realistic study, as shown in Figure 1. In Figure 1, the movable and deformable parts have been highlighted in red colour. The simulation of the other walls deformation would increase the calculation time, and the overall results should not vary much due to the rigidity of those parts. So their movement has not been modelled.

The full model was generated with Rhinoceros 5.0, a NURBS-based software. The epiglottis and glottis were modelled as three-dimensional warped surfaces without thickness. The geometrical details were taken from previous research studies (Friedrich and Lichtenegger 1997; Jotz et al. 2014). Additionally, special attention has been given to maintaining a realistic glottal-free section (Lagier et al. 2016).

The mesh density ensures grid-independent results. The mesh was created using the ANSYS Meshing pre-processing module. It is a fully structured mesh, in which the first cell size was adjusted to attain a $y+$ value of approximately 1. The near wall region was meshed with a layer of 15 linear-growth prismatic cells to solve the viscous region with precision. The total number of cells is approximately $2{\cdot}10^6$, see Figure 2.

2.2. Glottal motion

The glottis is composed of the vocal folds, the vocal process of the arytenoids, and the anterior and posterior commissures. The true vocal fold (TVF) movement during adduction and abduction has been measured with endoscopy by Dailey et al. (2005). Later, the TVF angle (β) has been measured during the cough from laryngeal videoendoscopic samples, digitalized at 30 Hz by Britton et al. (2012). The averages of the measures of the control patients taken by Britton et al. (2014) have been used in this study.

The movement of the TVF is simplified in Figure 3, such as the rotation (exemplified as adduction) of the point 0 to the point 1. It might appear to be easy to implement a rotation (centre of rotation point 3), where R is the radius, but this action would be possible only for points on the outer circumference (not in the TVF). Thus, the movement has been implemented as a rotation of the $\overline{4-0}$ segment, with respect to the centre of rotation (point 4), and by modifying properly this rotation, because it will move point 0–point 2. The airway circumference, as in Equation (1), and the equality of the slopes of the straight lines $\overline{4-1}$ and $\overline{1-2}$, as in Equation (2), have been used.

$$x_1^2 + y_1^2 + Ax_1 + By_1 + C = 0 \quad (1)$$

$$\frac{x_1 - x_4}{x_2 - x_4} = \frac{y_1 - y_4}{y_2 - y_4} \quad (2)$$

The parameters A, B, and C are shown in Equation (3).

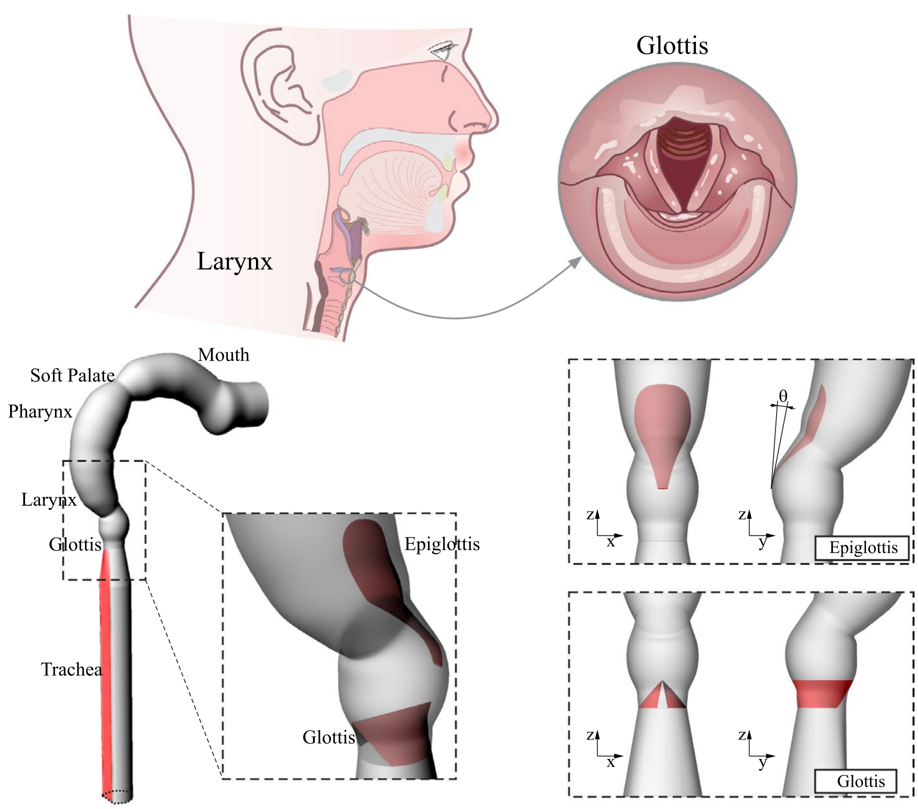

Figure 1. Synoptic diagram of the main areas that are involved in the cough maneuver and the geometry used in this study, including the glottis and epiglottis.

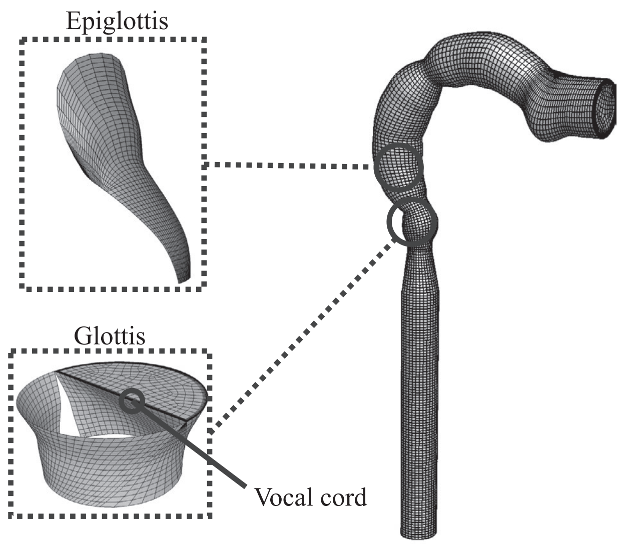

Figure 2. Mesh used, overview and details.

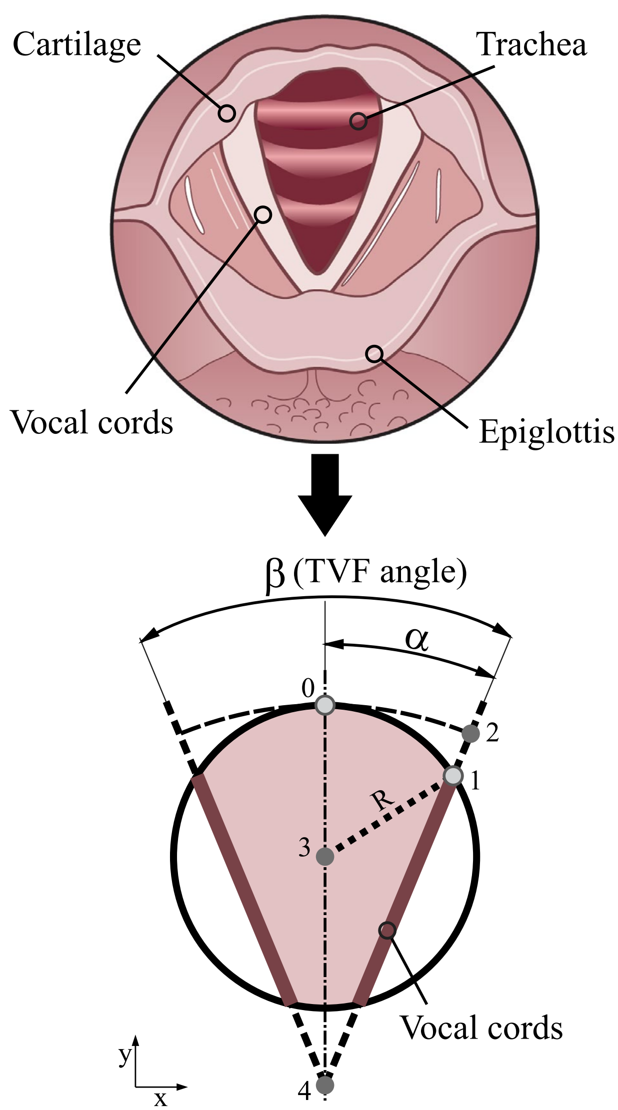

Figure 3. Cross-section of the larynx and details of the glottis simplification that is simulated.

$$A = -2x_3$$
$$B = -2y_3 \tag{3}$$
$$C = x_3^2 + y_3^2 + R^2$$

The rotation of the TVF has been extended linearly through all of the vestibular folds. In the top part of the vestibular fold, the rotation is equal to the TVF rotation, and it decreases linearly with the z coordinate, becoming zero (no rotation) in the bottom part.

2.3. Numerical modelling

The airflow conservation laws were solved together with the mucus film equations. We assumed a Newtonian, dry, and ideal gas for the mean airflow in the upper airways. The mass conservation law, Equation (4), and the momentum Equation (5) were solved.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{v}\right) = 0 \tag{4}$$

$$\rho \frac{\partial \vec{v}}{\partial t} + \rho\left(\vec{v} \cdot \nabla\right)\vec{v} = \rho \vec{g} - \nabla p + \mu \nabla^2 \vec{v} \tag{5}$$

where ρ is the air density, μ is the air viscosity, $\vec{v}$ is the air velocity vector, p is the air pressure, and $\vec{g}$ is the gravity vector.

The mucus layer structure has been simplified into a single equivalent layer, with a uniform thin initial thickness. The EWF model assumes that the mucus film is dominated by viscosity and the flow is nearly unidirectional. This assumption is made because the thickness of the mucus film is small in comparison to the radius of the curvature of the surface.

The mass conservation law applied to a two-dimensional mucus film in a three-dimensional domain is the differential Equation (6), where ρ_f is the mucus film density; z is the film height; ∇_S is the surface gradient operator; $\vec{v}_f$ is the mean film velocity; and $\dot{m}_S$ is the mass source per unit of wall area.

$$\frac{\partial z}{\partial t} + \nabla_S \cdot \left[z \vec{v}_f\right] = \frac{\dot{m}_S}{\rho_f} \tag{6}$$

Conservation of the film momentum is given in Equation (7), where the different pressures are defined in Equation (8).

$$\frac{\partial z \vec{v}_f}{\partial t} + \nabla_S \cdot \left[z \vec{v}_f \vec{v}_f\right] = -\frac{z \nabla_S p_f}{\rho_f} + \vec{g}_\tau z + \frac{3}{2\rho_f}\vec{T}_{fs} - \frac{3\mu_f}{z\rho_f}\vec{v}_f \tag{7}$$

$$p_f = p_{\text{gas}} + p_z + p_\sigma$$

$$p_z = -\rho z\left(\vec{n} \cdot \vec{g}\right)$$

$$p_\sigma = -\sigma \nabla_S \cdot \left(\nabla_S z\right) \quad (8)$$

where p_f is the film pressure, $\vec{g}$ is the gravity component normal to the surface, $\vec{T}_{fs}$ is the viscous shear force at the mucus-air interface, μ_f is the viscosity of the mucus film phase, p_{gas} is the gas pressure, ρ is the air density, $\vec{n}$ is the normal vector, and σ is the surface tension.

Mucus clearance was associated with different clearance mechanisms, such as shear force or pressure gradient, although there are others, such as transient wave formation (King et al. 1985). The differential equation for the film momentum, Equation (7), presents the three main mechanisms that can contribute to the mucus clearance process: the surface shear force, the pressure film gradient, and the surface tension. Each of these terms have been activated and deactivated properly to analyse their effect.

The fluid flow is affected by the changes in the deformable walls, but are not solved in a bi-coupled manner. The dynamic mesh motion and the pressure conditions were implemented as temporal functions in order to remain synchronized throughout the coughing process.

The simulated cough process begins with the inspiration phase and finishes at the end of the expulsion phase, passing for high velocity peaks and negligible airflow rates, and thus, flow regimes that range from laminar to clearly turbulent were expected. The turbulent model that suits best these conditions is the k–ω model, with the shear stress transport (SST) model enabled (Kleinstreuer and Zhang 2003). This model uses the k–ω model in the near wall region and switches to a k–ω formulation that is equivalent to the k–ε model in the outer region. This turbulent model was used successfully in airway studies (Mylavarapu et al. 2009; Sandeau et al. 2010). All of the simulations were calculated with a transient pressure-based solver performed in double precision. The SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) pressure-velocity coupling routine was used (Doormaal and Raithby 1984) with a least squares cell-based gradient, along with the QUICK scheme for the momentum and turbulence equations. The Eulerian wall film model equations were discretized with a second-order upwind scheme.

The simulations were conducted using an Intel® Xeon® Processor E5-2697 2.6 GHz cluster with 1280 GB of RAM. The model was solved using the CFD software Fluent Inc. ANSYS 17.1.

2.4. Boundary and initial conditions

A pressure outlet boundary condition was set to 0 Pa in the mouth exit. The pressure inlet condition has been imposed in the lower part of the trachea. The inlet pressure has been imposed as a temporal evolution that is parameterised from experimental measurements obtained from Yanagihara et al. (1966), as shown in Figure 4. The experimental curves were fitted to an equation that follows the same types of equations than Gupta et al. (2009) used for the flow rate.

Moreover, the pressure imposed at inlet should be the total pressure in order to achieve the measured flow rate. Thus, the dynamic pressure has been added to the static pressure temporal evolution, which was obtained with the instantaneous velocity. The velocity, in turn, was derived from the flow rate data. The total pressure obtained in this way has been included in Figure 4.

The rheological mucus properties were taken from a previous research study, in which the mucus apparent viscosity is calculated with a function of the air shear rate (Paz et al. 2017). The apparent viscosity of the mucus film layer was updated at each time step with the local values of the wall shear stress following the Equation 9.

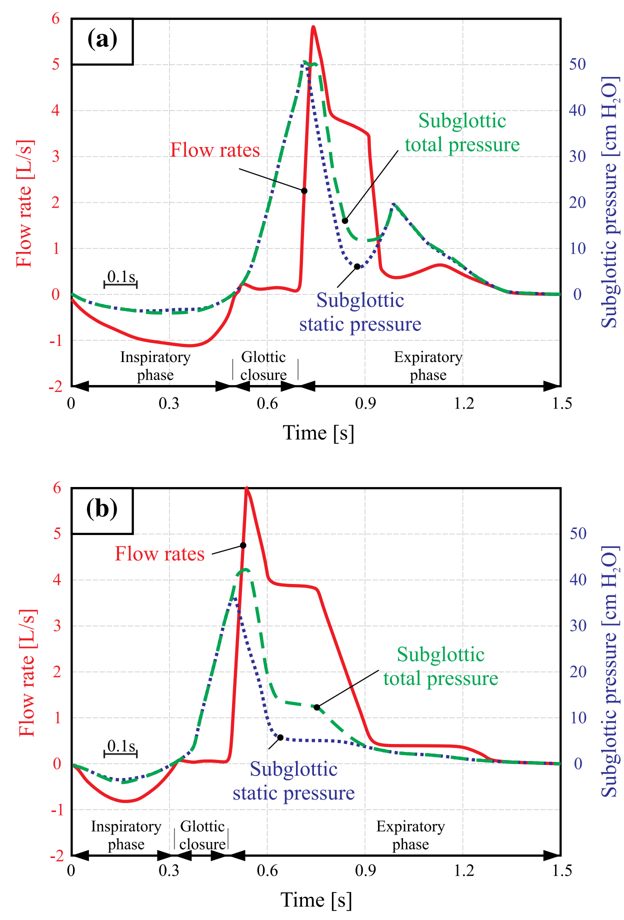

Figure 4. Flow rate and subglottic pressure, simultaneously recorded in a typical single cough. (a) with two subglottic pressure peaks in the positive pressure phase and (b) only one sharp peak in the positive pressure phase. Adapted from Yanagihara et al. (1966).

Where μ_{mucus}, is the mucus viscosity, τ_w, is the air shear stress, $\tau_{w\text{–min}}$ and $\tau_{w\text{–max}}$ are the minimum and maximum shear stress, respectively. Similarly, μ_{min} and μ_{max} are the minimum and maximum viscosity values, and a and b are the tuning parameters, taken from Paz et al. (2017).

$$\mu_{\text{mucus}} = \begin{cases} \mu_{\text{max}} & \tau_w < \tau_{w-\text{min}} \\ \mu = a(\tau_w)^b & \tau_{w-\text{min}} < \tau_w < \tau_{w-\text{max}} \\ \mu_{\text{min}} & \tau_w > \tau_{w-\text{max}} \end{cases} \quad (9)$$

The initial mucus thickness was fixed to 30 µm (Matsui et al. 1998), and the surface tension at 3.2·10-3 N/m (Hamed and Fiegel 2014). Zero velocities and pressures were assumed at all points to initialize the solution.

The airflow equations were solved using a time step that was fixed to 0.1 ms. The Eulerian film time step used was fixed at 0.01 ms. These values were optimised along the first simulations, thereby allowing us to have the scaled residuals of all of the variables below 10^{-3}, following the Fluent convergence criterion recommended (ANSYS 2015).

A smooth no-slip shear boundary condition was applied to all of the walls, but not all of the walls remained fixed during the simulation. The dynamic mesh method has been used to reproduce the trachea and glottis movement. The rear flexible membrane of the trachea followed the results of Williamson et al. (2011). The glottal motion has been previously explained, and the rotation equations have been implemented as User Defined Functions (UDF). There are two inputs needed to characterize correctly the glottal movement through the cough: the TVF angles and the angular velocities. It is known than during coughing, the glottis abduction is faster than the adduction, and this relationship is relevant for this research. Advances in laryngeal imaging allow us to measure the TVF angle during coughing and the average velocities of abduction and adduction from laryngeal videoendoscopic samples (Britton et al. 2012, 2014). The ranges of the maximum angles and mean velocities that were used in the simulations are shown in Figure 5(a) and (b), respectively.

To make a simpler and more intuitive implementation of glottis movement, we obtained the ranges of the times, as shown in Figure 5(c), from the values shown in Figure 5(a) and (b). These ranges represent the times spent in the abduction and adduction movements during a typical cough.

In all of these figures, the results are shown with a typical statistical boxplot representation, where the bottom and the top of the box represent the first and the third quartile, respectively (Q_1, Q_3). The lower and upper whiskers represent the maximum and the minimum, respectively. The line inside the box represents the median or second quartile, Q_2. The same criteria have been maintained in the results section.

For analysing a cough epoch, more parameters are needed to characterize the TVF movement. Figure 6 represents the general behaviour of a multiple cough process, with all of the times designated on the horizontal axis, and the sequence of corresponding TVF angles (β) represented on top. With all of these parameters, the dynamic mesh implementation is defined unambiguously. The reduction in the airflow shown in the second and subsequent

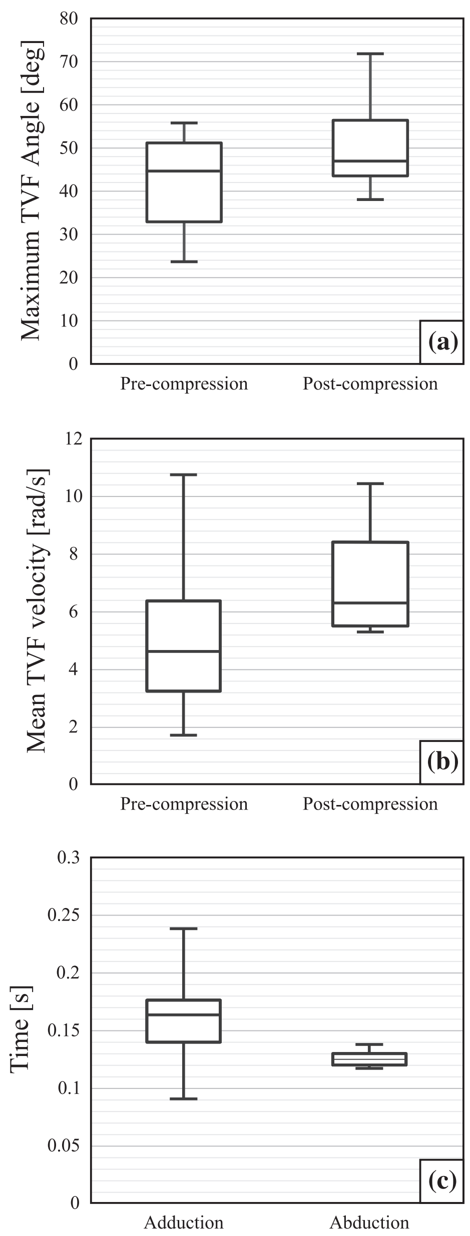

Figure 5. Maximum (a), and mean (b) true vocal fold (TVF) angle. Pre-compression = just prior to the pre-compression adduction; Post-compression = ±0.15 s from the peak expiratory cough flow, and (c) Time range for a normal healthy man. Obtained from the values of Britton et al. (2014).

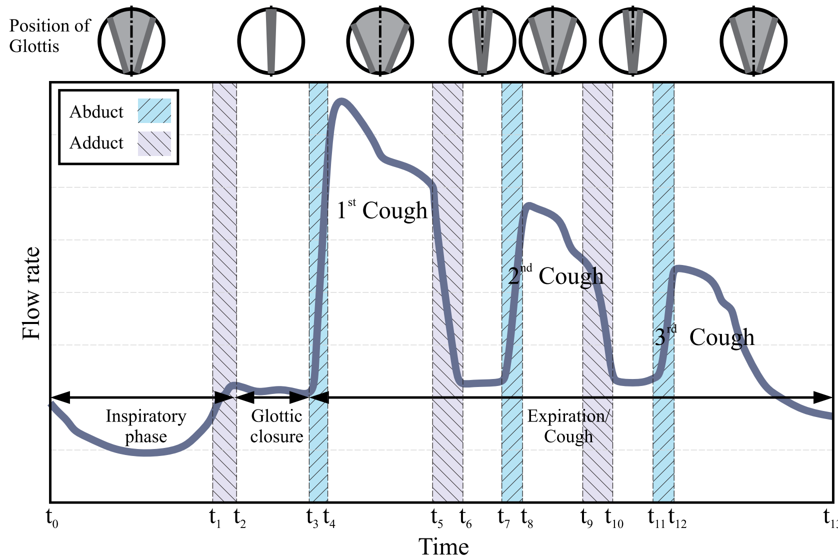

Figure 6. Simplification of the cough flow rate in a general cough epoch and the glottis position during the process.

coughs was implemented following the Gupta et al. (2009) correlation, which was experimentally obtained from 25 healthy patients whose sequential coughs were measured.

3. Results and discussion

A total of 386 cases were simulated with the transient methodology, where the full range (min to max) of the parameters of healthy people were analysed. These simulations include the range of TVF angles and velocities, cough assistance device tests, various epiglottis positions, single and consecutive coughs, and different clearance mechanisms being active.

In all of the simulated cases, the temporal evolution of the air flow rate, pressures, velocities, mucus mass, and mucus thickness, and the areas of all of the moving parts of the model during the cough cycle, were stored as text files and subsequently analysed in MATLAB. With the mucus layer thickness evolution, the cough clearance efficiencies were evaluated following the definition of Equation (10) for each case.

$$\text{Cough Efficiency (CE)} = \frac{\text{Removed mucus mass}}{\text{Mucus mass before cough}} \cdot 100 \tag{10}$$

3.1. Glottis effects

The obtained cough efficiency (CE) ranges obviously depend on physiological values being used as inputs, and these are the most important parameters that affect the CE. However, maintaining the same 'normal' or 'average healthy inputs', the effect of the glottis movement can by analysed. The dynamic simulations, which account for the abduct and adduct movements of the glottis, capture more details and offer more realistic results than the simulations without the movements. The maximum air velocity increases from 60 to 190 m/s considering the glottis movement, and the mean air velocity increases by 5%. Due to the proposed dynamic mesh implementation, it is possible to analyse the glottis closure time during the cough, but also the glottis abduction and adduction times. This advance means that the simulation of the larynx movements enabled us to study the process in depth.

Assuming that a glottal closure time, ($t_3 - t_2$), is zero, to avoid introducing the closed time factor, the glottis adduction has been advanced in tests 1 to 11, as shown in Table 1. The results show that the CE is almost unaffected by the time adduction star as long as the inspiration phase is sufficiently effective. There is no clear tendency with regard to this parameter.

The effect of the glottis adducted angle (β_1 in Table 1) has been evaluated by simulating from 0 to 44° in tests from 12 to 20, as shown in Table 1. The trend is clear: the smaller the value of β_1 (more closed the glottis) is, the higher the value of CE. However, the larger the angle is, the smaller the effect. With these results, the effect of the glottis closure on the CE is quantified. The reduction of CE is important, but the minimum CE magnitude, practically without glottis movement (for average healthy person parameters) can still be considered to be an effective cough.

The glottis closure time (t_3 − t_2) has been simulated from 0 to 0.5 s (0.05 s step). As long as the magnitude of the pressure peak and the exhaust air flow are maintained in the same order, the efficiency of the cough remains in almost the same range of values. The CE peak (44.6%) was reached for a closure time of 0.2 s, and the range of the CE variation was lower than −3%.

Table 1. Summary of some tests, where the times [s] are given, β is the TVF angle [deg], and the cough efficiency is (CE). The sub-index means: 0-precompression, 1-clossure, 2-postcompression.

Tests	t_0	t_1	t_2	t_3	t_4	β_0	β_1	β_2	CE (%)
1	0	0.23	0.31	0.31	0.4	44.5	0	47	41.80
2	0	0.215	0.295	0.295	0.385	44.5	0	47	41.16
3	0	0.2	0.28	0.28	0.37	44.5	0	47	40.60
4	0	0.185	0.265	0.265	0.355	44.5	0	47	40.53
5	0	0.17	0.25	0.25	0.34	44.5	0	47	39.77
6	0	0.155	0.235	0.235	0.325	44.5	0	47	39.50
7	0	0.14	0.22	0.22	0.31	44.5	0	47	41.01
8	0	0.125	0.205	0.205	0.295	44.5	0	47	39.52
9	0	0.11	0.19	0.19	0.28	44.5	0	47	40.56
10	0	0.095	0.175	0.175	0.265	44.5	0	47	40.07
11	0	0.08	0.16	0.16	0.25	44.5	0	47	39.48
12	0	0.23	0.31	0.31	0.4	44.5	5	47	41.35
13	0	0.23	0.31	0.31	0.4	44.5	10	47	38.38
14	0	0.23	0.31	0.31	0.4	44.5	15	47	37.51
15	0	0.23	0.31	0.31	0.4	44.5	20	47	36.61
16	0	0.23	0.31	0.31	0.4	44.5	25	47	34.96
17	0	0.23	0.31	0.31	0.4	44.5	30	47	33.88
18	0	0.23	0.31	0.31	0.4	44.5	35	47	33.31
19	0	0.23	0.31	0.31	0.4	44.5	40	47	33.01
20	0	0.23	0.31	0.31	0.4	44.5	44	47	32.68

Moreover, in the single cough study, the location of the glottis closure movement, with respect to the pressure and flow rate evolution, has been simulated. It could be said that the glottis motion curve has 'slipped' over the time, whereas the pressure and flowrate curves remain fixed over the time. There are some diseases that affect the coordination of the glottis movement, such as the paradoxical vocal fold movement (PVFM; Hartley et al. 2015; Rosen et al. 2016), which can be considered to be similar to the simulated conditions. The slip range was simulated from 0 to 0.2 s. All of the simulated tests show a notable reduction in the CE, which is usually greater as the slip increases. A phase shift of 0.1 s reduces the CE to 12.18%, and with the extreme slip values, the CE becomes almost zero.

A second airflow rise is common in the cough, which is usually associated with an increase in the subglottal pressure and, thus, is most likely a consequence of a second partial closure of the glottis.

More than 50 tests of a second, third, and fourth cough (consecutive) have been simulated, using the air flow reduction parameters of Gupta et al. (2009). The second cough improves the CE appreciably (5–15% in the full cough range parameters), but the third cough implies an insignificant improvement (less than 0.5% in all tests), and the fourth coughs did not change the CE (always employing a reduction in the Gupta et al. parameters for an average man).

The tests with different compression times for the second cough ($t_7 - t_6$), in the same range as before, have almost not affected the global CE. The effect of the glottis adducted angle in the second cough has been evaluated with the same range as in a single cough, from 0 to 44°. The behaviour in these tests shows a lower reduction in the CE with an increase in the β_1. Thus, it could be said that in the second cough, the trends are very similar to the first cough, but in general, the second cough has less weight than the first cough on the CE, and therefore, the effects of this changes in the second cough are of less magnitude.

3.2. Assistance cough devices

Assistance coughing techniques are commonly used in patients with neuromuscular diseases, due to the dysfunction in the respiratory muscles (Torres-Castro et al. 2014; Kim et al. 2016). They are also commonly used in tracheotomised patients or different respiratory dysfunctional patients. The cough assistance devices are connected to the patient, typically via an oronasal mask, and periodically give inspiratory and expiratory pressure.

These devices increase the cough flow and provide help in the clearance process, but in spite of being clinically proven, it is complex to quantify their effect. Due to the quantification of the CE, which allows the developed model, it is possible to analyse the effect of the operating range of the cough assistance devices. The pressure ranges in the cough assistance devices are approximately 40 cm H_2O for the inspiratory pressure and −40 cm H_2O expiratory pressure (Boitano 2006; Kim et al. 2016). Some researchers (Whitney et al. 2002) recommend starting with lower pressures, approximately 25/−30 cm H_2O, and increasing them to 40/40 cm H_2O. The values of 40/−40 cm H_2O are considered to be adequate for reaching a peak of expiratory flow sufficient for an effective cough (Bach 1993). It must be clarified that this analysis has not reproduced the operation of any specific commercial device.

The pressure that was added to reproduce the effect of cough assistance devices has the range of −4000 to +4000 Pa, step 500 Pa. Other factors that could be associated with these diseases, such as modifications of the mucus viscosity and thickness, have not been considered with regard to obtaining comparable results.

The improvement in the CE as associated with cough assistance devices is compared in a visual way, with the after-cough mucus thickness contours of Figure 7-top. In this figure, the first, second, and third quartiles of healthy people cough values (Britton et al. 2014) are compared with and without cough assistance, using ±4000 Pa.

$$\text{Cough Efficiency Increase (CEI)} = \frac{\text{CE}_{\text{New}} - \text{CE}_{\text{Initial}}}{\text{CE}_{\text{Initial}}} \cdot 100 \tag{11}$$

To quantify the effect of the used operation pressure, the CE increase (CEI) rate defined in Equation 11 has been compared; see Figure 7-bottom. The CEI increases with the pressure with a logarithmic growth ($R = 0.9937$), and

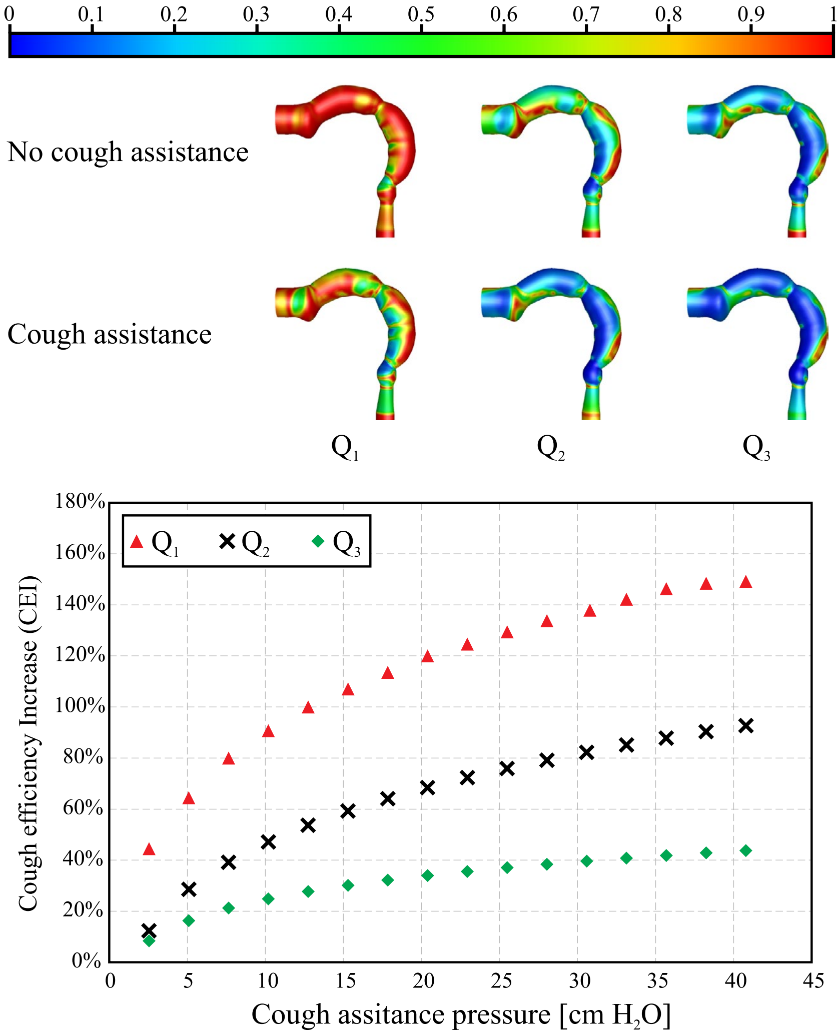

Figure 7. Effect of a cough assistance device. Top: Dimensionless mucus thickness using +4000/−4000 Pa. Bottom: Cough efficiency increase rate at different operating pressures.

thus, with moderate values of pressure increases, the CEI values that are reached are quite high.

3.3. Epiglottis effect

The presence of an epiglottis in a healthy subject does not affect the CE significantly (the range of θ simulated was from 0 to 15°). Nevertheless, the airflow changes with the epiglottis position; see Figure 8. These different behaviours imply changes in the pressure field and shear stress force distribution and, also, changes in the mucus thickness distribution. There are not common pathologies associated with epiglottis movement or position causing cough, although there are some diseases that are associated with the production of the cough reflex in the epiglottis, which are related to chronic cough due to enlarged epiglottis (Gurgel et al. 2008) or abnormal uvula length (Loftus et al. 2016).

3.4. Cough clearance mechanisms

The differential equation for the film momentum, Equation 7, presents the three main mechanisms that can contribute to the mucus clearance process: the viscous shear force, the gradient of the film pressure, and the surface tension. The contribution of each of these mechanisms to the CE has been analysed in a wide range of cough parameters, which correspond to the full range of a cough registered in normal healthy men (Britton et al. 2014). The CE results obtained in the simulations use different combinations of clearance mechanisms and are shown in Figure 9(a).

The surface tension affects the interfacial instability and the formation of droplets, removing the mucus mechanisms. However, the low thickness/radio ratio and the high Weber number values during the coughing process reduce its contribution (Moriarty and Grotberg 1999; Hasan et al. 2010). The surface tension term contribution to the CE is very small, comparing (III) and (IV),

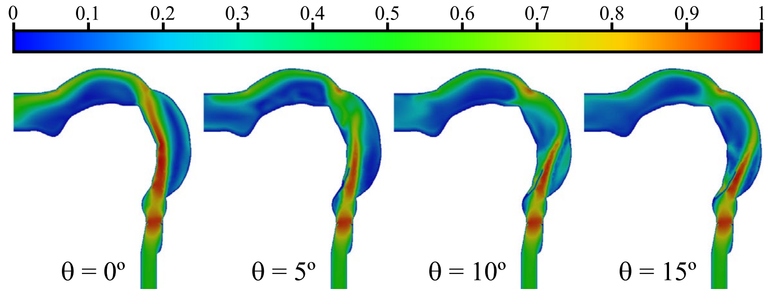

Figure 8. Dimensionless velocity (with u_{ref} = 60 m/s) in the mean plane of the airway, using four epiglottis angles during the cough process.

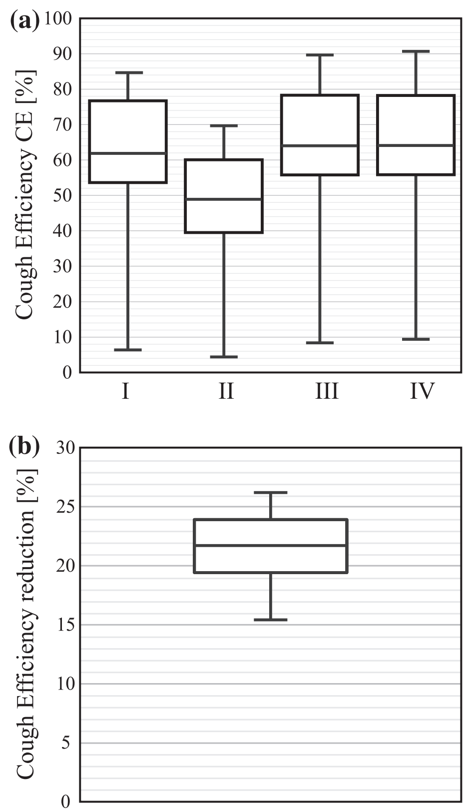

Figure 9. Boxplots of the CE results obtained by EWF simulations, using the full range of healthy subjects, from Britton et al. (2014). (a) The mechanisms of the clearance actions were the following: (I) Viscous shear force. (II) Gradient of film pressure. (III)Viscous shear force and gradient of film pressure. (IV) Viscous shear force and gradient of film pressure and surface tension and (b) Boxplot of the CE reduction from the viscous shear force to the gradient of the film pressure.

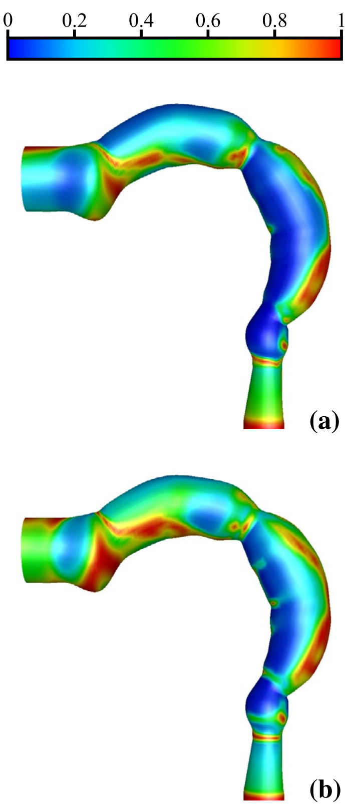

Figure 10. Dimensionless mucus layer thickness after the cough process with the mechanism of clearance activity. (a) Viscous shear force and (b) Gradient of film pressure.

and could be considered to be negligible for the mucus rheology values employed in this research.

The 0.5 Mach number is exceeded in the peak flow passing through the glottis, and therefore, the air's compressibility might affect the results.

The film pressure gradient term, and especially the viscous shear force term, are cited in most publications as being responsible for the cough clearance effect; thus, those should be considered to be dominant. When the terms of the viscous shear force and gradient pressure are simultaneously present, as in Figure 9(a)-III, the contribution of the shear force is larger than that of the gradient pressure. Analysing the contribution of each of them separately, simulating individually (I) and (II) in Figure 9(a), it is shown that the CE with the shear force is significantly greater than that with the gradient of the film pressure.

The reduction in the CE is shown in the boxplot of Figure 9(b); it is approximately 20%. With this significant but not dramatic reduction, the clearance process can be considered to be effective even without the viscous shear force contribution.

Moreover, the reduction is shown in the contours of the final mucus thickness, as shown in Figure 10, where there are appreciable differences also in the locations of the minima and maxima values due to the clearance mechanism's distinctive behaviour. These differences in the thickness show that the cough clearance process is subject to several mechanisms, the pressure gradient and the shear force, which have the same order of magnitude, and it is a complex phenomenon.

4. Conclusions

The cough clearance process has been simulated with an EWF model that includes the vocal fold movement due to using the dynamic mesh method. This accomplishment has enabled us to analyse the impact of the glottis movement on the CE. The glottis closure improves substantially the CE results; nevertheless, even with a fixed glottis, the cough can still be considered to be effective. Other parameters, such as the glottis closure time or the abduction and adduction velocity, have been found to be not relevant in the CE.

Moreover, the differences of a single cough and a cough epoch have been quantified due to the proposed methodology. The cough assistance devices effect on the CE has been reproduced appropriately. The CEI has logarithmic growth with the operating pressure. The epiglottis position in the analysed range does not have a significant effect on the CE. The viscous shear force is the main mechanism in the cough clearance process, but the gradient film pressure has a substantial effect and should not be neglected. However, these results should be validated in a more quantitative manner, with *in vivo* data. The proposed methodology has allowed the analysis of a complex phenomenon, which considers many different boundary conditions, with a reasonable computational cost.

Disclosure statement

No potential conflict of interest was reported by the authors.

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